

MIDTERM I

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Instructions. The exam has a total of 120 points. I will grade it over 100. The points of each question is shown in boldface. Please use this information to allocate your time efficiently. Start writing your answers in the answer sheet provided but you can use as much paper as needed. The answer sheet has page number 1, so be sure to number any extra page that you use consecutively. Write your answers neatly and cross anything that you don't want me to consider. At the end of the exam I will staple all your papers.

1. **(30 points.)** *Choice under Uncertainty.* Suppose that X is a set of prizes, and that $x \in X$ denotes a prize. Let also Ω denote the set of states and that $s \in \Omega$ denotes a state. For each state s , and lottery f , define

$$B(s, f) = \{x \in X : f(x|s) \geq f(x'|s), \forall x' \in X\}.$$

Thus, $B(s, f)$ denotes the prize (or prizes) that has (have) the *highest likelihood* according to lottery f at state s . The notation $|B(s, f)|$ denotes the cardinality of $B(s, f)$, i.e. the number of its elements. A decision-maker (DM) says that, if he knew that the true state of the world was in some set T , then we would weakly prefer a lottery f over another lottery g ($f \succeq_T g$) if and only if

$$\sum_{s \in T} |B(s, f)| \geq \sum_{s \in T} |B(s, g)|.$$

- (a) **(6 points.)** Interpret the condition that defines the attitude of the decision maker regarding lotteries.

- (b) **(12 points.)** Do the DM's preferences satisfy the Continuity Axiom?

Axiom (Continuity). If $f \succeq_S g$ and $g \succeq_S h$, then $\exists \gamma \in [0, 1]$ and a lottery g such that $g \sim_S \gamma f + (1 - \gamma)h$.

- (c) **(12 points.)** Do the DM's preferences satisfy the Objective Substitution Axioms?

Axiom (Objective Substitution and Strict Objective Substitution). If $e \succeq_S f$ and $g \succeq_S h$ and $\alpha \in [0, 1]$, then $\alpha e + (1 - \alpha)g \succeq_S \alpha f + (1 - \alpha)h$.

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2. **(70 points + extra 20 points.)** *Procurement Contracts.* A government offers a firm a military contractor to buy $q \geq 0$ units of military hardware, in exchange for a payment of $t \geq 0$. The contractor's constant unit cost is private information to it and is either high (c_h) or low (c_ℓ) with $c_h > c_\ell$. The government's beliefs about the military contractor's cost is given by $\Pr(c_\ell) = p \in (0, 1)$. If the firm with cost c_i , $i = h, \ell$, accepts the contract (t_i, q_i) , its profits is $t_i - c_i q_i$, whereas if it rejects it receives zero profit. The payoff for the government is given by

$$\ln(q) - t.$$

- (a) **(20 points.)** Assume that the government observes the firm's type. Find the First Best Contract.

- (b) **(30 points.)** Now assume that the government cannot observe the firm's type. Find the optimal contract. Compare with the first best.

- (c) **(20 points.)** Assume that now the government faces a contractor with *three possible* types: high, medium and low ($c_h > c_m > c_\ell$). The beliefs of the government regarding these costs are: $\Pr(c_\ell) = p_\ell \in (0, 1)$, $\Pr(c_m) = p_m \in (0, 1)$ and $\Pr(c_h) = p_h \in (0, 1)$. Characterize as much as you can the optimal contract in this case.

- (d) **(20 points.)** *This one is harder.* Recall that monotonicity of the allocation is an important consequence of incentive compatibility. In the solution that you found in part (c), can you find conditions on the probabilities and/or the costs such that the solution is monotonic? If the conditions you propose fail, what would you do to alter the solution such that it is monotonic?